💻 Take-Home Technical Assessment: E-Commerce Scraper

⏳ Time Estimate: 1-2 Hours

We value your time, so if you find yourself spending longer than two hours, please stop and submit your work as-is, noting where you ran out of time. The goal is to see your *approach* to application development and *code quality* rather than a fully optimized solution.

🛠️ The Challenge

Your task is to create a **command line application** that scrapes a publicly available e-commerce test page, processes the product data, and outputs a single JSON structure.

1. Target URL

The page to scrape is:

https://webscraper.io/test-sites/e-commerce/static

2. Requirements

The application must perform the following steps and adhere to the output structure:

- **Web Scrape:** Fetch the main page and follow **all** available product links.
- **Data Aggregation:** Collect the following data for every available product:
  - `name` (string)
  - `description` (string)
  - `price` (number, e.g., $1178.19 → 1178.19)
  - `colors` (string array, *required only if multiple colour options are present*)
- **HDD/Configuration Handling:** If a product has multiple HDD (storage) options (e.g., 128GB, 256GB), **each option must be returned as a separate product** in the final `results` array, with the name updated to reflect the specific configuration.
- **Output Format:** The application must output a single JSON object containing a `results` array and a `total` field.

3. Output Specification

The final output must be a single JSON object with two top-level fields:

1. **`results`**: A JSON array containing all individual products (including separated HDD options).
2. **`total`**: The sum of all unique product prices in the `results` array.

Example JSON Structure (for illustration):

JSON


```
{
    "results": [
        {
            "name": "Dell Latitude 5580 128 GB",
            "description": "15.6\" FHD, Core i5-7300U, 16GB, 256GB SSD",
            "price": 1178.19
        },
        {
            "name": "Samsung Galaxy",
            "description": "5 mpx. Android 5.0",
            "price": 93.99,
            "colors": ["gold", "white", "black"]
        }
    ],
    "total": 1272.18
}
```


💡 Guidance & Freedom

- **Language & Libraries:** You may use **any programming language, tools, or libraries** you prefer.
- **AI Tools: We explicitly encourage the use of AI assistants** (like Claude Code, GitHub Copilot, ChatGPT, or Gemini) to generate boilerplate code, suggest solutions, or look up documentation. This is a realistic part of modern development.
- **Best Practices:** Please ensure your submission reflects best-practice application development. This is your opportunity to demonstrate your knowledge, experience and opinions. No one line perl scripts!

📦 Submission

Please submit your solution by sending a link to the source code (e.g., a public or unlisted GitHub repository) to **steven@natter.co**.

We look forward to reviewing your submission!